Cybersecurity Hands-On Lab 1

Module 1: Understanding Basic Security Weaknesses

Part 1: Understanding Password Security

Step 1: Analyze Weak Passwords

Examples of very common weak passwords: 123456 password admin 12345678 qwerty

Let us discuss the following questions:

• Why are these passwords insecure?

These passwords are insecure for several key reasons:

1. Too Common
   Millions of users use the same passwords.
   Attackers already have lists of the most common passwords (from data breaches).
2. Predictable patterns
   "123456" and "qwerty" follow simple keyboard or numeric sequences.
   "admin" and "password" are default or obvious choices.
3. Too short
   Short passwords have fewer possible combinations.
   This makes them easier to guess.
4. No complexity
   They lack:
   - Uppercase letters
   - Special characters
   - Randomness

In short: they are easy to guess because they are predictable and widely known.

• How quickly could a computer guess these passwords?

Computer can guess Very fast and often instantly.
With modern computers:
Common passwords like 123456 can be guessed in less than a second
Even slightly longer ones like 12345678 still cracked almost instantly
Why:
Attack tools don't guess randomly at first, they:

Try known common passwords first
Use optimized algorithms and massive datasets
Result: These passwords are usually cracked in milliseconds to seconds

Attackers often use automated tools that perform:

• dictionary attacks

It is the uses of a list of common words and passwords
Example list :
password
admin
welcome
Very effective against weak passwords. Fast and efficient

• brute force attacks

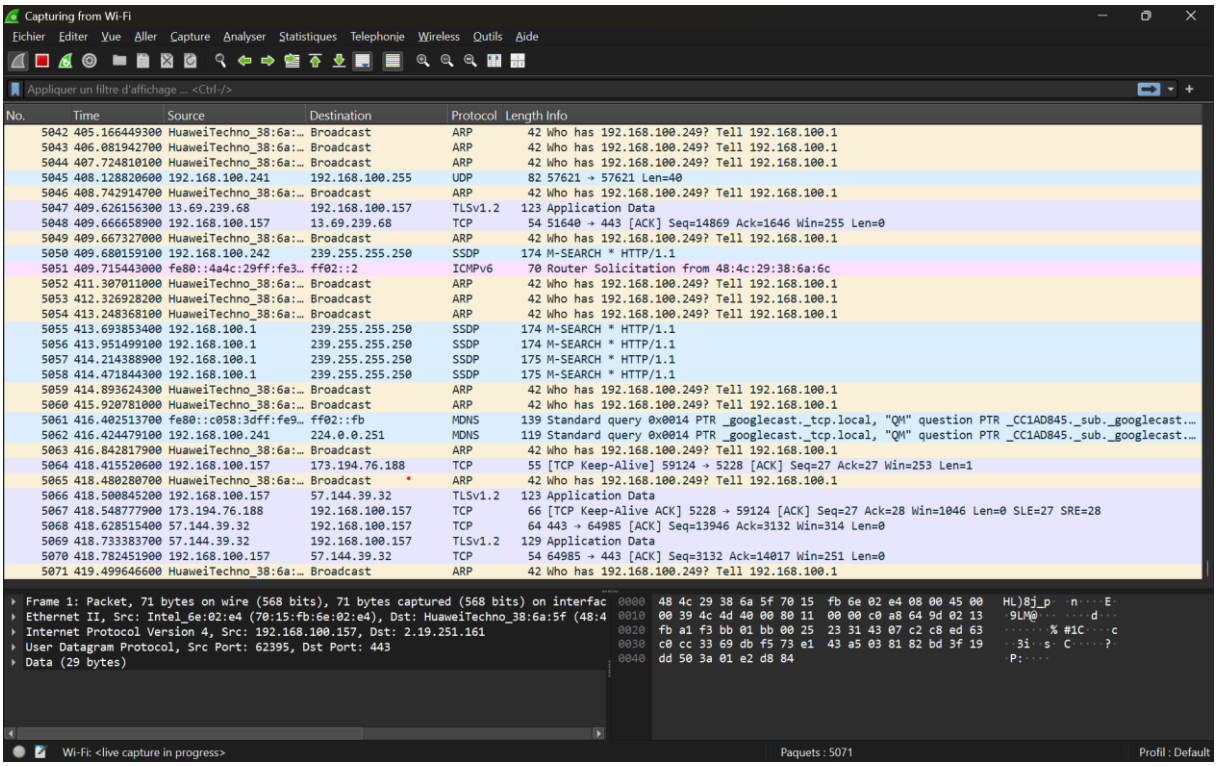
It consists to tries every possible combination
Example :
aaaa → aaab → aaac → ...
Slower than dictionary attacks, but guaranteed to succeed eventually, faster if the password is short

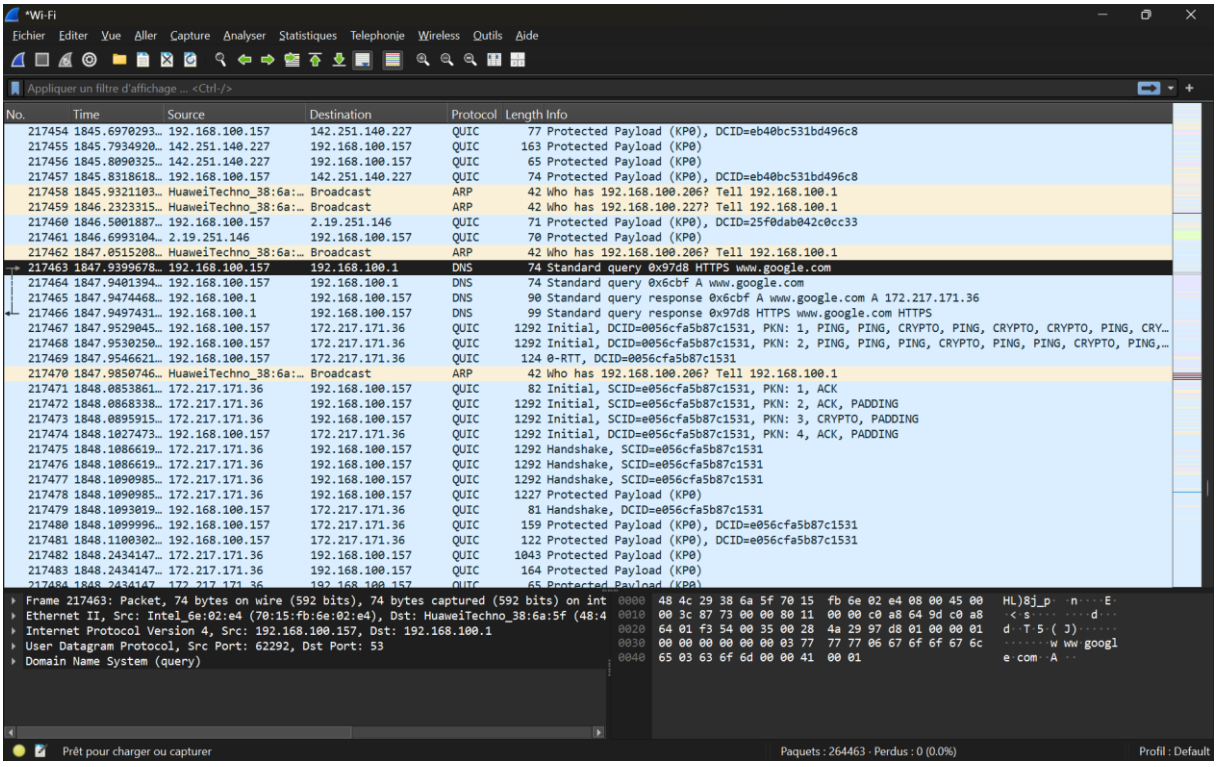
Step 2: Create Strong Passwords

Let us create three strong passwords following these rules:
- minimum 12 characters
- mixture of uppercase and lowercase letters
- include numbers
- include special characters

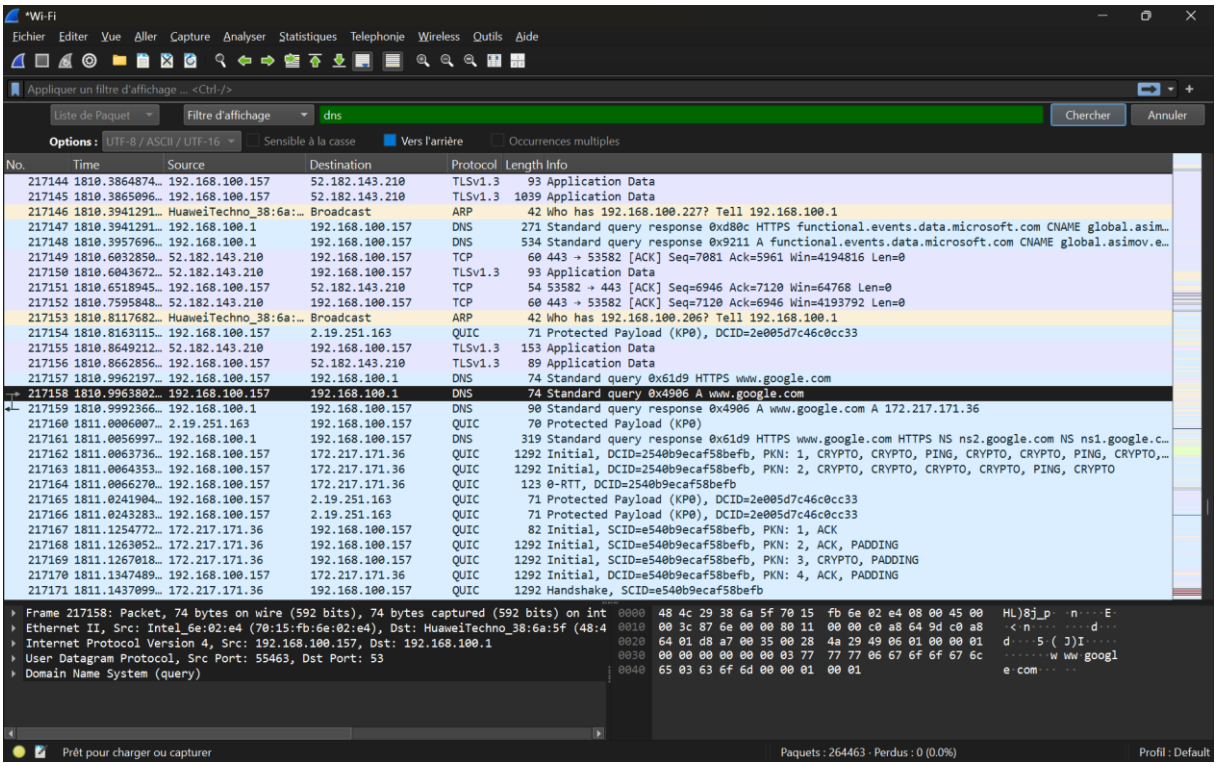
1…… T9!vR2#kLm8@QxP
2…… Z@7mN!4pL#s2Wq8X
3…… F3$uK!9rT@6yB#dL

Part 2 : Observing Network Traffic

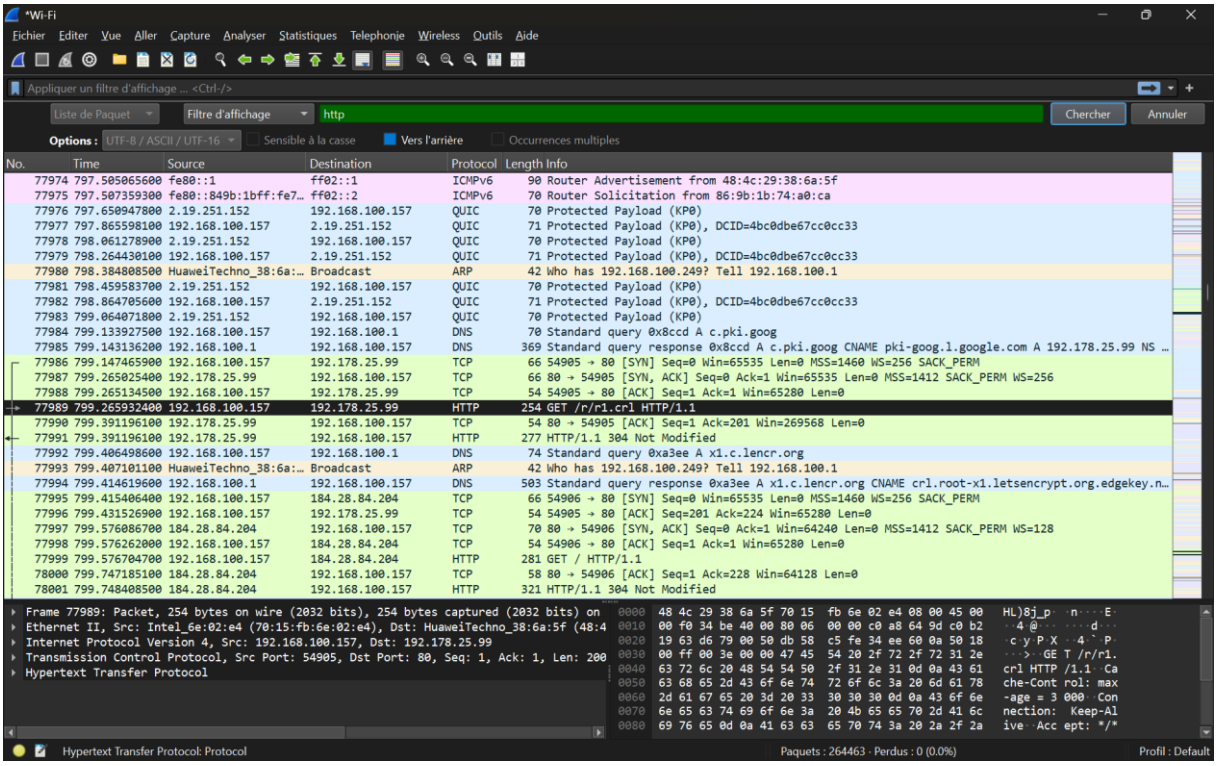


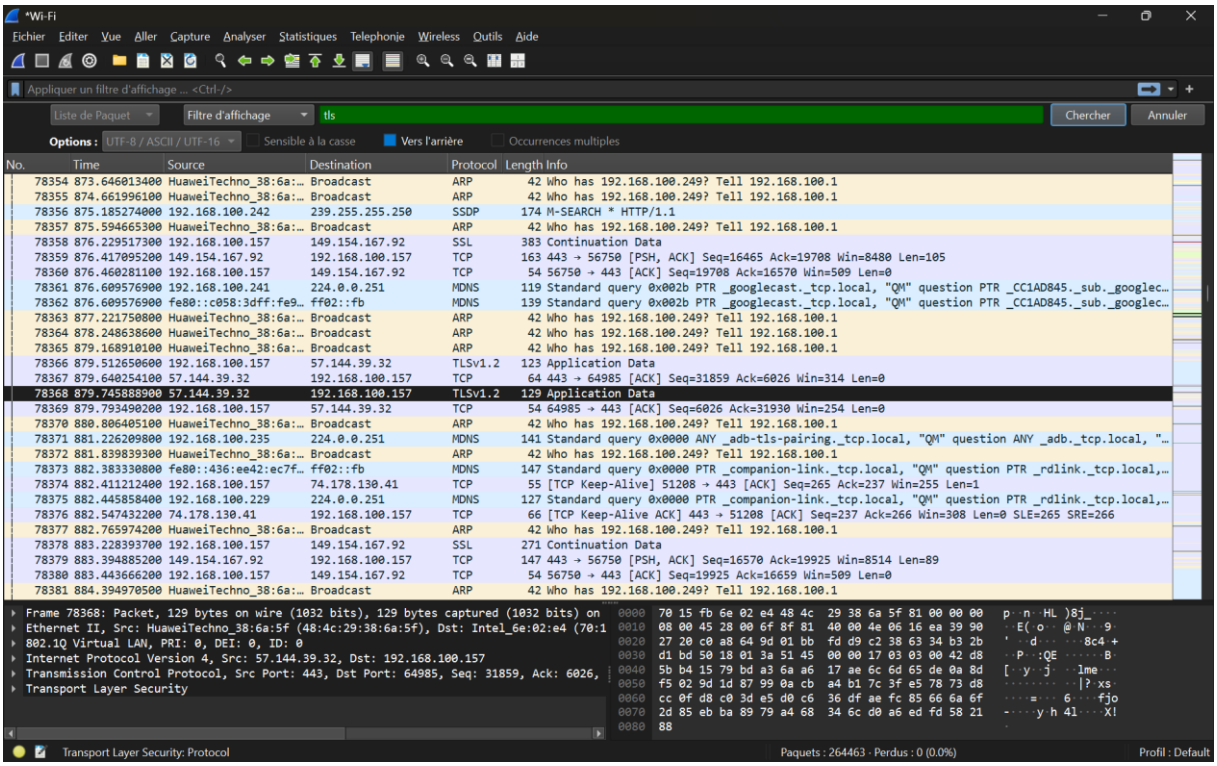


Part 3 : Packet Analysis



Task: Write one domain name observed: www.google.com





Analysis Questions

Answer the following questions.

1. Approximately how many packets were captured?

Approximately 264,463 packets were captured.

2. List three protocols observed during the experiment.

DNS
TCP
HTTP

3. Why is encryption important when browsing the internet?

Encryption is important because it protects data from being read by unauthorized users.

It ensures :

- Confidentiality (data is hidden)
- Security (prevents interception)
- Privacy (protects personal information)

Without encryption, anyone on the network can see your data.

4. What type of information could attackers obtain from unencrypted traffic?

Attackers could obtain sensitive information such as:

- Visited URLs/websites
- Usernames and passwords
- Form data (login details, messages)
- Cookies and session data
- Browser and device information